

CASCADIA DEPARTMENT OF TRANSPORTATION

Office of ITS Engineering & Procurement

ORIGINAL ISSUE

REQUEST FOR PROPOSAL

Statewide Intelligent Transportation System (ITS) Modernization & Traffic Management Center Upgrade Program

A multi-year, multi-phase statewide ITS infrastructure and software modernization program

RFP Number	RFP-CDOT-2026-1187
Issue Date	January 8, 2026
Proposal Submission Deadline	February 20, 2026, 4:00 PM PT
Issuing Organization	Cascadia Department of Transportation (CDOT)
Category	Intelligent Transportation Systems — Statewide ITS & TMC Modernization
Estimated Contract Value	See Section 12 — Commercial Terms

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1. Introduction & Background

The Cascadia Department of Transportation (“CDOT” or “the Department”) operates and maintains a statewide Intelligent Transportation System (ITS) network spanning approximately 1,450 centerline miles of interstate and state highways. The network includes 620 closed-circuit television (CCTV) cameras, 340 dynamic message signs (DMS), and 180 ramp metering locations, all connected to a Traffic Management Center (TMC) located in Cascadia City.

CDOT's current Advanced Traffic Management System (ATMS) software platform was deployed in 2011. The platform vendor has announced that extended support will end in 2027, and the platform no longer supports modern integrations, AI-based incident detection, or current cybersecurity standards. CDOT is issuing this Request for Proposal (RFP) to solicit qualified vendors to modernize its ATMS platform, refresh field device hardware across the network, and upgrade the communications backbone connecting field devices to the TMC.

This program will be delivered in three phases: a Pilot Corridor phase, a Regional Rollout phase, and a Statewide Rollout phase, as described in Section 9 (Project Timeline).

2. Definitions

Term	Definition
ITS	Intelligent Transportation System — the combined field devices, communications network, and software used to monitor and manage highway traffic.
ATMS	Advanced Traffic Management System — the central software platform used by TMC operators to monitor and control field devices.
TMC	Traffic Management Center — CDOT's central facility for statewide traffic monitoring and incident response.
DMS	Dynamic Message Sign — an electronic roadside sign used to display traveler information and alerts.
CCTV	Closed-Circuit Television — roadside cameras used for traffic monitoring and incident verification.
Ramp Meter	A traffic signal on a freeway on-ramp used to regulate the rate of vehicles entering the freeway.
511 System	CDOT's statewide traveler information phone and web service.
NIST	National Institute of Standards and Technology — issuer of the cybersecurity framework referenced in this RFP.
SLA	Service Level Agreement — a contractually defined performance commitment.
MP	Mile Post — the reference marker system used to identify highway locations.

3. Project Objectives

- Replace the legacy ATMS platform with a modern, standards-based ITS software platform.
- Refresh field device hardware (CCTV, DMS, ramp meters) that has reached or is approaching end of life.
- Upgrade the communications backbone connecting field devices to the TMC to support higher-bandwidth video and data.
- Implement AI-based automated incident detection to reduce average incident response time by at least 20%.
- Integrate the ATMS platform with CDOT's statewide 511 traveler information system.
- Harden ITS network cybersecurity in alignment with the NIST Cybersecurity Framework.

4. Scope of Work

1. Design, procure, and deploy a replacement Advanced Traffic Management System (ATMS) software platform.
2. Replace 620 end-of-life CCTV cameras with new pan-tilt-zoom (PTZ) IP cameras across all corridors listed in Appendix E.
3. Replace or upgrade 340 dynamic message sign (DMS) controllers, retaining existing sign structures where feasible.
4. Modernize 180 ramp metering locations with adaptive, demand-responsive metering algorithms.
5. Upgrade the fiber and wireless communications backbone connecting all field devices to the TMC.
6. Integrate the new ATMS platform with CDOT's statewide 511 traveler information system.
7. Implement AI-based automated incident detection algorithms across all CCTV-covered corridors.
8. Provide TMC operator training for approximately 45 operators across three shifts.
9. Harden ITS network cybersecurity in alignment with the NIST Cybersecurity Framework.
10. Provide 5 years of post-deployment operations and maintenance (O&M) support.

5. Technical Requirements

5.1 ATMS Software Platform

- Support for NTCIP 1200-series standards for field device communication.
- Map-based operator console showing real-time status of all field devices.
- Configurable incident management workflow with automated response plan suggestions.
- Role-based access control aligned to TMC operator, supervisor, and administrator roles.

5.2 CCTV Cameras

- Minimum 1080p resolution with 30x optical zoom.
- Integrated video analytics for stopped-vehicle and wrong-way-driver detection.
- IP66-rated enclosures rated for the Department's operating climate range.

5.3 Dynamic Message Signs (DMS)

- Full-matrix LED display, minimum 3-line message capability.
- Solar-compatible power option for remote sign locations.
- NTCIP 1203 compliant sign controllers.

5.4 Ramp Metering

- Adaptive metering algorithms responsive to real-time mainline volume and occupancy.
- Local override capability for TMC operators during incidents.

5.5 Communications Network

- Fiber-optic backbone with a minimum 10 Gbps core ring capacity.
- Cellular or licensed-wireless backup connectivity for remote field device locations.
- Network redundancy such that no single fiber cut disables more than 5% of connected field devices.

5.6 Hosting, Availability & Security

- Deployment model: CDOT-hosted on-premise at the TMC, or vendor-hosted cloud environment — vendor may propose either model with justification.
- System uptime SLA of 99.5% or higher for the ATMS software platform, measured monthly.
- Single sign-on (SSO) support via SAML 2.0, integrated with CDOT's existing Active Directory.
- Encryption at rest (AES-256 or equivalent) and in transit (TLS 1.2 or higher).
- Video and event data retention of a minimum of 2 years.

5.7 Cybersecurity

- Network architecture aligned with the NIST Cybersecurity Framework (Identify, Protect, Detect, Respond, Recover).
- Network segmentation isolating ITS field device traffic from CDOT's corporate IT network.
- Annual third-party penetration testing of the ATMS platform and communications network.

5.8 System Integrations

- Integration with CDOT's statewide 511 traveler information system for automated DMS message posting.
- API-based data feed to CDOT's public open-data portal for anonymized traffic volume data.

- Integration with the Department's existing computer-aided dispatch (CAD) system for incident data exchange.

6. Functional Requirements Matrix

Vendors must indicate compliance with each functional requirement in their proposal response using the codes: F (Fully Compliant), P (Partially Compliant — explain), C (Customization Required), N (Not Supported).

ID	Requirement	Priority
FR-01	Real-time map-based device status console	Mandatory
FR-02	Automated incident detection from CCTV video analytics	Mandatory
FR-03	Configurable incident response plan library	Mandatory
FR-04	Automated DMS message posting from approved response plans	Mandatory
FR-05	Adaptive ramp metering with manual override	Mandatory
FR-06	Historical event and video data archive with search	Mandatory
FR-07	Role-based dashboard customization	Desired
FR-08	Mobile app for field technician device diagnostics	Desired
FR-09	Automated device health monitoring with predictive maintenance alerts	Desired
FR-10	Public open-data API for traffic volume exports	Mandatory
FR-11	511 system integration for automated traveler alerts	Mandatory
FR-12	CAD system integration for incident data exchange	Mandatory

7. Eligibility Criteria

To be eligible for consideration, vendors must meet all of the following minimum criteria:

- Minimum of 8 years of experience providing ITS/ATMS systems integration services to state or provincial transportation agencies.
- Minimum annual revenue of \$40,000,000 (audited financials for the most recent fiscal year required).
- At least 4 reference implementations for state or provincial DOT ITS deployments covering 500 or more field devices, completed within the past 7 years.
- A Professional Engineer (PE) licensed in the state, on staff, responsible for ITS system design.
- Minimum insurance coverage: \$10,000,000 general liability; \$5,000,000 professional liability/E&O.
- Performance bond equal to 100% of the awarded contract value.

8. Deliverables

#	Deliverable	Due
1	Project Charter & Implementation Plan	45 days after NTP
2	ATMS Platform System Design Document	90 days after NTP
3	Pilot Corridor Deployment (I-90, MP 0-15)	Per Phase 1 milestone
4	Pilot Corridor Performance Report	60 days after Phase 1 go-live
5	Regional Rollout Deployment Plan	Prior to Phase 2
6	Regional Rollout Deployment (5 corridors)	Per Phase 2 milestone
7	Statewide Rollout Deployment Plan	Prior to Phase 3
8	Statewide Rollout Deployment (remaining corridors)	Per Phase 3 milestone
9	Cybersecurity Architecture & Penetration Test Report	Prior to each phase go-live
10	TMC Operator Training Materials & Delivered Sessions (45 operators)	Prior to each phase go-live
11	511 & CAD Integration Testing Report	Prior to statewide go-live
12	Final Documentation & As-Built Package	Project close-out

9. Project Timeline

Milestone	Date
RFP Issue Date	January 8, 2026
Vendor Q&A Deadline	January 22, 2026
Proposal Submission Deadline	February 20, 2026
Shortlist Notification	March 6, 2026
Vendor Presentations	March 16 – 20, 2026
Contract Award	April 6, 2026
Notice to Proceed (NTP)	April 27, 2026
Phase 1 — Pilot Corridor Go-Live (I-90)	February 2027

Milestone	Date
Phase 2 — Regional Rollout Go-Live (5 corridors)	November 2027
Phase 3 — Statewide Rollout Go-Live (remaining corridors)	August 2028
Full Project Completion	December 2028

10. Evaluation Criteria

Proposals will be scored on a 100-point scale as follows:

Criterion	Weight
Technical Solution & Architecture	25 points
Vendor Experience & References	20 points
Implementation Approach & Timeline	15 points
Cost	25 points
Cybersecurity Approach	10 points
O&M / Support Plan	5 points

11. Insurance & Bonding Requirements

- Commercial General Liability: \$10,000,000 per occurrence.
- Professional Liability / Errors & Omissions: \$5,000,000 per claim.
- Performance Bond: 100% of total contract value, issued by a surety rated A- or better by A.M. Best.
- CDOT and its officers, agents, and employees must be named as additional insureds on all policies except professional liability.

12. Commercial Terms

Estimated Project Budget: \$85,000,000 – \$110,000,000

Contract Term: 6 years, with two (2) optional 12-month renewals.

Payment Schedule

- 10% upon contract signing / NTP
- 15% upon System Design Document acceptance
- 20% upon Phase 1 Pilot Corridor go-live
- 25% upon Phase 2 Regional Rollout go-live
- 20% upon Phase 3 Statewide Rollout go-live
- 10% upon final acceptance

Other Commercial Terms

- Warranty period: 3 years following each phase's go-live, covering defect remediation at no additional cost.
- Payment terms: Net 45 from invoice date.
- Liquidated damages: 0.25% of total contract value per week of delay against the approved schedule, capped at 8% of total contract value.

13. Submission Instructions

Proposals must be submitted electronically in PDF format to its-procurement@cdot.state.gov no later than the deadline listed above. Proposals must remain valid for 150 days from the submission deadline. Late submissions will not be considered.

14. Terms & Conditions

14.1 Governing Law

This RFP and any resulting contract shall be governed by the laws of the State of Cascadia.

14.2 Confidentiality

All proposal materials submitted become the property of CDOT and are subject to applicable public records law, subject to any properly marked trade secret exemptions.

14.3 Reservation of Rights

CDOT reserves the right to reject any or all proposals, waive minor irregularities, and negotiate final contract terms with the selected vendor.

14.4 Intellectual Property

All custom configurations, integrations, and documentation developed specifically for CDOT under this engagement shall become the property of CDOT upon final payment. Vendor retains ownership of pre-existing, generally licensed software.

14.5 Indemnification

Vendor shall indemnify, defend, and hold harmless CDOT and its officers, agents, and employees from any claims, damages, or losses arising from the vendor's negligent acts or omissions in the performance of this contract.

14.6 Dispute Resolution

The parties shall attempt to resolve any dispute through good-faith negotiation before pursuing formal legal remedies. Unresolved disputes shall be subject to the jurisdiction of the courts of the State of Cascadia.

14.7 Force Majeure

Neither party shall be liable for delays caused by events beyond its reasonable control, including natural disasters, acts of government, or other force majeure events, provided the affected party gives prompt notice and uses reasonable efforts to mitigate the delay.

14.8 Assignment

Vendor may not assign or subcontract a material portion of this contract without CDOT's prior written consent.

15. Contact Information

Issuing Officer	Angela Turlington
Title	ITS Procurement Manager
Email	aturlington@cdot.state.gov
Phone	(360) 555-0173

Appendix A: Pricing Form

Vendors must complete the following pricing form in full. Unit prices must be all-inclusive of labor, materials, travel, and overhead unless otherwise noted.

Line Item	Description	Price
A-1	ATMS Software Platform Licensing (Statewide)	\$_____
A-2	ATMS Implementation & Configuration Services	\$_____
A-3	CCTV Camera Hardware & Installation (620 units)	\$_____
A-4	DMS Controller Hardware & Installation (340 units)	\$_____
A-5	Ramp Metering Upgrade (180 locations)	\$_____
A-6	Communications Backbone Upgrade	\$_____
A-7	511 & CAD Integration Services	\$_____
A-8	Cybersecurity Architecture & Penetration Testing	\$_____
A-9	TMC Operator Training (45 operators)	\$_____
A-10	5 Years Post-Deployment O&M Support	\$_____
A-11	Total Proposed Price	\$_____

Appendix B: Reference Form

Provide at least 4 references meeting the eligibility criteria in Section 7. Duplicate this table as needed.

Field	Response
Client Agency	
Contact Name & Title	
Phone / Email	
Device Count / Corridor Miles	
Contract Value	
Completion Date	

Appendix C: Compliance Certification Checklist

Requirement	Certified (Y/N)
Vendor meets minimum 8 years relevant experience	
Vendor meets minimum annual revenue threshold	
Vendor holds required insurance coverage	
Vendor can provide performance bond as specified	
Vendor's ITS design lead is a licensed PE	
Proposed ATMS supports NTCIP 1200-series standards	

Requirement	Certified (Y/N)
Proposed CCTV cameras meet minimum resolution/zoom specification	
Proposed solution aligns with the NIST Cybersecurity Framework	

Appendix D: Technical Specifications Detail

D.1 CCTV Camera Specification

Parameter	Minimum Requirement
Resolution	1080p (Full HD)
Optical Zoom	30x
Enclosure Rating	IP66
Low-Light Performance	0.01 lux color, 0.001 lux B/W
Video Analytics	Stopped-vehicle and wrong-way detection
Communication Protocol	ONVIF Profile S/T compliant

D.2 Dynamic Message Sign (DMS) Specification

Parameter	Minimum Requirement
Display Type	Full-matrix LED
Message Lines	3 lines minimum
Brightness	Auto-dimming, 1,000–10,000 nits range
Controller Protocol	NTCIP 1203 compliant
Power Option	AC and solar-compatible

D.3 Ramp Meter Controller Specification

Parameter	Minimum Requirement
Metering Algorithm	Adaptive, volume/occupancy responsive
Detector Interface	Loop and radar detector compatible
Override Capability	Local and remote TMC override
Communication Protocol	NTCIP 1207 compliant

D.4 Communications Network Specification

Parameter	Minimum Requirement
Core Backbone Capacity	10 Gbps minimum, ring topology
Field Device Connectivity	Fiber preferred; licensed wireless where fiber infeasible
Redundancy	No single fiber cut disables more than 5% of devices
Backup Connectivity	Cellular failover for remote sites

Appendix E: Corridor & Device Inventory List

The following table lists the major highway corridors included in the scope of this program, with existing device counts to be replaced or upgraded. Minor connector routes and short spur segments are not itemized individually here but are included in the statewide totals in Section 1; a complete site-by-site inventory will be provided to the selected vendor during the design phase. Vendors should use this list to develop phased deployment plans and cost estimates.

Corridor ID	Route	Segment Description	CCTV Cameras	DMS Units	Ramp Meters
C-01	I-90	MP 0–15, Cascadia City – Eastgate	18	9	12
C-02	I-90	MP 15–42, Eastgate – Summit Pass	14	7	0
C-03	I-5	MP 0–22, Port Neelan – Cascadia City	26	14	20
C-04	I-5	MP 22–58, Cascadia City – Fairhaven	31	16	24
C-05	I-5	MP 58–90, Fairhaven – Northgate Border	19	11	8
C-06	I-405	MP 0–17, Eastgate Loop	22	12	18
C-07	US-2	MP 0–35, Cascadia City – Timberline	12	6	0
C-08	US-2	MP 35–71, Timberline – Highpass	9	5	0
C-09	SR-522	MP 0–19, Northgate – Riverside	15	8	10
C-10	SR-167	MP 0–24, Valley Junction – Southport	17	9	14
C-11	I-82	MP 0–38, Southport – Vinewood	13	7	0
C-12	US-97	MP 0–45, Highpass – Dryland	8	4	0
C-13	SR-14	MP 0–28, Riverside – Gorgeview	11	6	0
C-14	I-205	MP 0–12, Southport Loop	16	9	12
C-15	SR-99	MP 0–16, Cascadia City Waterfront	20	11	0

Total: 15 corridors — 251 CCTV cameras, 134 DMS units, 118 ramp meters.

Appendix F: Sample Service Level Metrics

The following illustrative service levels will inform the O&M support contract to be negotiated with the selected vendor following statewide go-live:

Metric	Target
ATMS platform uptime	99.5% monthly
Critical field device fault response (on-site)	Within 8 business hours
Non-critical field device fault response (on-site)	Within 3 business days
Software defect (Severity 1) resolution	Within 24 hours
Software defect (Severity 2) resolution	Within 5 business days